

1. The triangular-wave voltage source is applied to the circuit in Fig. 1(a). The triangular-wave voltage is shown in Fig. 1(b). Estimate the average power delivered to the resistor $R = 80\Omega$ when the circuit is in steady-state operation. (Expand v_g up to 3rd harmonic term). [9]

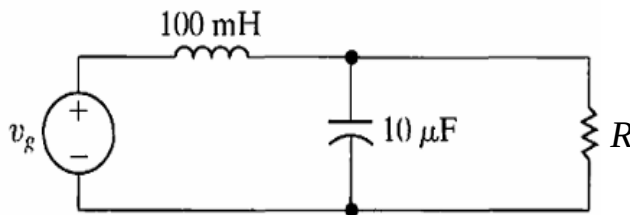


Fig.1(a)

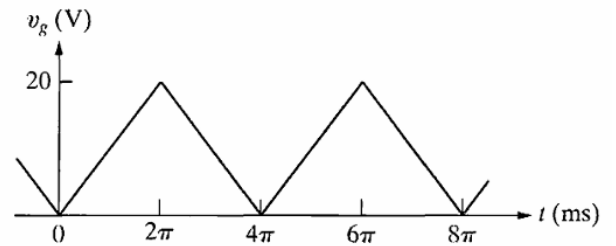


Fig.1(b)

2. The following dc measurements were made on the resistive network shown in Fig. 2. $R = 260\Omega$

Measurement 1

$$V_1 = 4\text{ V}$$

$$I_1 = 5\text{ mA}$$

$$V_2 = 0\text{ V}$$

$$I_2 = -200\text{ mA}$$

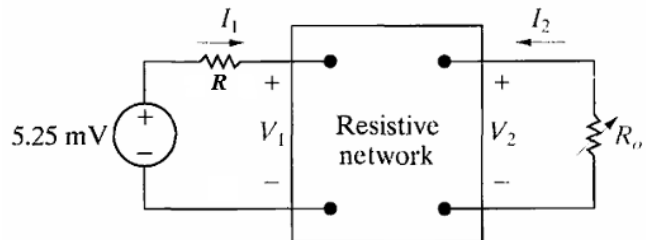
Measurement 2

$$V_1 = 20\text{ mV}$$

$$I_1 = 20\text{ μA}$$

$$V_2 = 40\text{ V}$$

$$I_2 = 0\text{ A}$$



A variable resistor R_0 is connected across port 2 and adjusted for maximum power transfer to R_0 . Find the maximum power. [6]

$$Z_{Th} = z_{22} - \frac{z_{12}z_{21}}{z_{11} + Z_g}$$

$$Z_{Th} = \frac{1 + y_{11}Z_g}{y_{22} + \Delta y Z_g}$$

$$Z_{Th} = \frac{a_{12} + a_{22}Z_g}{a_{11} + a_{21}Z_g}$$

$$Z_{Th} = \frac{Z_g + h_{11}}{h_{22}Z_g + \Delta h}$$